

A Transdiagnostic Investigation of Pavlovian-to-Instrumental Transfer and Its Neurocognitive Underpinnings



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Background

Pavlovian-to-Instrumental Transfer (PIT)

Pavlovian conditioned stimuli (CS) can influence the accuracy & intensity of instrumental behavior¹

pos. CS □ activation + approach

neg. CS □ inhibition + avoidance



- Increased in preclinical^{2,3} & clinical^{4,5} AUD samples
- Associated with increases in alcohol consumption³ & relapses⁵
- Neural correlates in NAcc², amygdala⁶, IPFC² & dmPFC²
- Heterogeneous (↔⁷) or no evidence for other substances

Aim 1:

Test the transdiagnostic nature of PIT

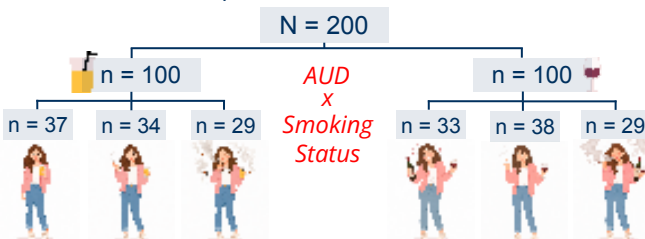
Aim 2:

Investigate the neurocognitive underpinnings of PIT

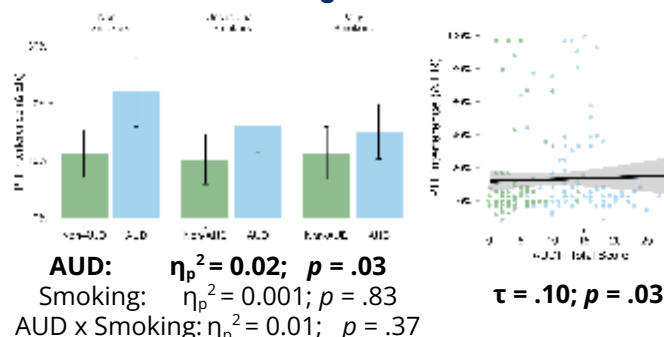
Methods



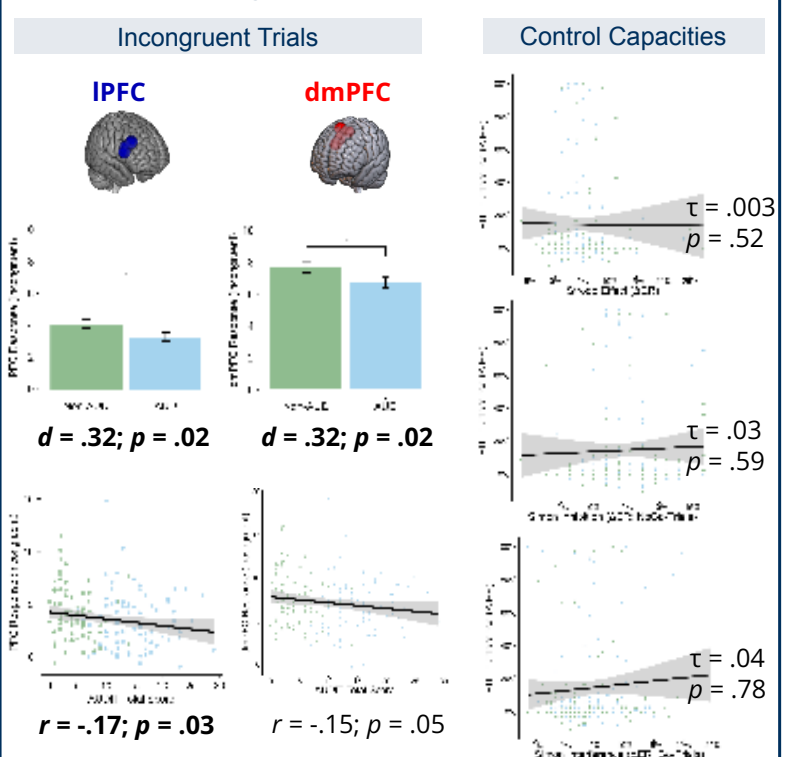
Sample Stratification at BA



Results: AUD x Smoking Status

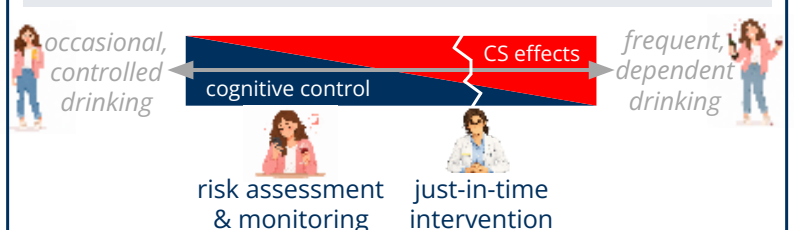


Results: PIT & Cognitive Control



Conclusion & Anne's case

- 1 Increased PIT interference across AUD spectrum
- 2 No evidence for similar effects in smoking or interaction
- 3 Potential link with reduced recruitment of control resources



Publications

Thill, L., Chen, H., Baeuchl, C., Krasniqi, S., Born, M., Heinz, A., & Smolka, M.N. (in Prep.). *Reduced Control Over Pavlovian-to-Instrumental Transfer Conflicts in Alcohol Use Disorder But Not Smoking.*

Baeuchl, C., Chen, H., Thill, L., Heinz, A., & Smolka, M.N. (2025). Erhöhter alkohol-spezifischer Pawlowsch-instrumenteller Transfer bei Alkoholkonsumstörung. *Deutscher Suchtkongress*, 2(1), 2292. <https://doi.org/10.18416/DSK.2025.2292>

Rapp, M., Wiedenhoft, J.L., ..., Thill, L., ..., Heinz, A., & Banaschewski, T. (under review). Impulsivity, loneliness, and stress in individuals with alcohol use disorder across the lifespan. *Submitted to Journal of Studies on Alcohol and Drugs.*